

1 Student-t Likelihood Derivation

1.1 Input and Assumption Nodes

Input node: `model.student_t_noise`.

1.2 Derivation

With residual $r_i = y_i - x_i^\top \beta$, the likelihood is the product of Student-t densities:

$$p(y \mid X, \beta, \sigma, \nu) = \prod_{i=1}^n \frac{\Gamma((\nu+1)/2)}{\Gamma(\nu/2)\sqrt{\nu\pi}\sigma} \left(1 + \frac{r_i^2}{\nu\sigma^2}\right)^{-(\nu+1)/2}.$$

1.3 Result

The result node is `likelihood.student_t`.